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Making pens from scratch – part 1

In the first part of a new project, **Walter Hall** turns the core section of a fountain pen

Not even the most dedicated of pen turners could argue that making simple pens from kits is the most challenging aspect of the hobby of woodturning. In previous articles in this series I have looked at several ways in which the developing maker might seek out more challenging methods and techniques for working with kits or making their own blanks. Eventually, however, the more adventurous amongst pen makers will seek out the ultimate challenge of making pens from scratch, using only basic components such as nibs and feed mechanisms for fountain pens and refills and click or twist mechanisms for ballpoints.

Over the next few articles I shall be making some pens from scratch, starting this month with the core of a fountain pen, usually known as the 'section'. This is the part of the pen that holds the nib and the ink feed or cartridge together and is threaded to fit into the body of the pen. I made this one, for reasons I shall explain later, from aluminium. Next month I shall complete the pen by making the body (barrel) and cap.

I should explain at the outset that I am not an expert in making pens from scratch, and I shall be sharing my learning process and disasters with you along with my successes. Those with more experience of making pens from scratch or anyone with ideas or techniques is welcome to contact me with suggestions. I will be happy to incorporate or share them as appropriate in future articles.

Making the 'section'

The principle skills involved in making this part of the pen are rather more akin to precision engineering than woodturning. Accurate measurement, careful drilling and tapping and the cutting of external threads with dies are at the core of the process. Some specialist tools will be needed – principally correct sized drills, fine taps and dies and the means to use them on or off the lathe. Finding some of the more specialist taps and dies in the UK can be a challenge, I had to import some of the ones I used from the USA where they are more readily available.

A tailstock mounted die holder is also essential to ensure that externally threaded components have threads that are straight and true to the axis of the component.

Choosing an appropriate material for the section is also vital. The research I have done suggests that one of the best materials for making the section of a pen is ebonite, which is a kind of hard rubber. Other suitable options are alumilite resin or soft metals such as aluminium or brass. Some polyester resins, as I found from experience, are too brittle for the making of such delicate parts. I shall conduct more experiments with other materials later and am happy to receive suggestions.

The choice of nib, feed and ink converter will determine the dimensions and tap sizes required. I used a 5mm Bock nib unit from Beaufort Ink who also supplied the ink converter and the appropriate tap for the feed. This, and the availability of suitable taps and dies led me to decide on 8.5mm x 0.75 threads for the section to barrel connection and a triple start 11mm thread for the cap to barrel, which will be used in the next article. These latter taps and dies all had to be imported. You may wish to try the more affordable and easily obtainable standard metric fine threads until you see if pen making from scratch is for you.

WALTER HALL

Walter Hall is a woodturner who has specialised in making pens and pencils for more than 20 years. Based on the beautiful Northumberland coast in the UK, Walter sells his bespoke pens and pencils through local craft centres and via his website.

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The nib unit came from Beaufort Ink

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